

Let f be a polynomial of degree at least 2.

If w is an attractive fixed point of f , we define the "basin of attraction of w " as

$$A(w) = \{z \in \mathbb{C} : f^k(z) \rightarrow w \text{ as } k \rightarrow \infty\}$$

The "basin of attraction of infinity" is defined as

$$A(\infty) = \{z \in \mathbb{C} : |f^k(z)| \rightarrow \infty \text{ as } k \rightarrow \infty\}$$

Note that $A(w)$ and $A(\infty)$ are open in $\mathbb{C}$.

Prop: Let w be an attractive fixed of f or $w = \infty$.
Then $\partial A(w) = J(f)$.

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Mandelbrot set

For any $c \in \mathbb{C}$, let $f_c(z) = z^2 + c$.

We define the Mandelbrot set M as follows:

$$M = \{c \in \mathbb{C} : J(f_c) \text{ is connected}\}$$

Let $h(z) = \alpha z + \beta$, where $\alpha \neq 0$.

$$\begin{aligned} h^{-1} \circ f_c \circ h(z) &= h^{-1}((\alpha z + \beta)^2 + c) \\ &= h^{-1}(\alpha^2 z^2 + 2\alpha\beta z + \beta^2 + c) \\ &= \frac{\alpha^2 z^2 + 2\alpha\beta z + \beta^2 + c - \beta}{\alpha} \\ &= \alpha z^2 + 2\beta z + \frac{\beta^2 - \beta + c}{\alpha} \end{aligned}$$

$\therefore$ For any quadratic polynomial $f(z)$, we can find α, β & c s.t. $f(z) = h^{-1} \circ f_c \circ h(z)$.

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Since $f = h^{-1} \circ f_c \circ h$, $f^k = h^{-1} \circ f_c^k \circ h$.

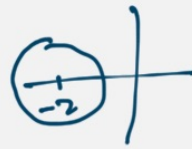
This implies $f^k(z) \rightarrow \infty$ iff $f_c^k(h(z)) \rightarrow \infty$

$$\therefore J(f) = h^{-1}(J(f_c))$$

Example: Find the Julia set of $f(z) = z^2 + 4z + 2$

$$f(z) = (z+2)^2 - 2$$

If $h(z) = z+2$, then $f(z) = h^{-1} \circ f_0 \circ h(z)$,
where $f_0(z) = z^2$

Since, $J(f_0) = \{z : |z|=1\}$, 

$$J(f) = h^{-1}(\{z : |z|=1\})$$

$$= \{z-2 : |z|=1\}$$

$$= \{w : |w+2|=1\} = \text{circle centered at } -2 \text{ with radius } 1$$

Theorem: The Mandelbrot set M is

$$M = \{c \in \mathbb{C} : J(f_c) \text{ is connected}\}$$

$$= \{c \in \mathbb{C} : \{f_c^k(0)\}_{k=1}^{\infty} \text{ is bounded}\}$$

$$= \{c \in \mathbb{C} : f_c^k(0) \not\rightarrow \infty \text{ as } k \rightarrow \infty\}$$

Exercise 14.5 $f(z) = z^2 - 2$. Find $J(f)$.

First, let's find the fixed points of f .

$$f(z) = z \Leftrightarrow z^2 - z - 2 = 0$$

$$\Leftrightarrow (z-2)(z+1) = 0$$

$$\Leftrightarrow z = 2 \text{ or } z = -1$$

$$f'(z) = 2z, \quad |f'(2)| = 4 > 1, \quad |f'(-1)| = 2 > 1.$$

$\therefore 2$ & -1 are both repelling fixed pts. of f .

$$\Rightarrow \{2, -1\} \subseteq \partial(f)$$

$$1) \quad J(f) = \overline{\bigcup_{k=1}^{\infty} f^{-k}(2)} = \overline{\bigcup_{k=1}^{\infty} f^k(-1)}$$

$$f^{-1}(2) = \{\pm 2\}$$

$$f^{-2}(2) = \{0, \pm 2\}, \quad f^{-3}(2) = ?, \dots$$

Claim: $J(f) \subseteq [-2, 2]$ (why?)

$$f(0) = -2, \quad f^2(0) = f(-2) = 2,$$

$$f^3(0) = f(2) = 2, \quad f^k(0) = 2 \quad \forall k \geq 2.$$

$$\Rightarrow \{f^k(0)\}_{k=1}^{\infty} \text{ is bounded.}$$

$$\Rightarrow c = -2 \in M$$

$$\Rightarrow J(f) \text{ is connected.}$$

Since $J(f) \subseteq [-2, 2]$ and $-2, 2 \in J(f)$, we get
 $J(f) = [-2, 2]$

Try up to Exercise 14.12 in Chapter 14
 of Falconer.